PART 3: BERT-LSTM Hybrid Sentiment Analysis

PACKAGE IMPORTS

The primary objective of this project is to build a hybrid deep learning classifier model that can predict the sentiment of IMDB movie reviews. The dataset contains a collection of movie reviews and each review is labeled as either positive or negative. Using text representation, hybrid deep learning architecture (bert-lstm or PT-LSTM) and various classification algorithms, the project will aim to develop a model that can effectively classify the sentiment of movie reviews. The model's performance will be evaluated using standard classification metrics, such as accuracy, precision, recall, and F1-score.

```
!pip install -q -U keras-nlp tensorflow-text
```

```
Requirement already satisfied: keras-nlp in /usr/local/lib/python3.11/dist-packages (0.21.1)
Requirement already satisfied: tensorflow-text in /usr/local/lib/python3.11/dist-packages (2.19.0)
Requirement already satisfied: keras-hub==0.21.1.dev202505120324 in /usr/local/lib/python3.11/dist-packages (0.21.1.dev202505120324)
Requirement already satisfied: keras-core in /usr/local/lib/python3.11/dist-packages (0.1.7)
Requirement already satisfied: keras-tuner in /usr/local/lib/python3.11/dist-packages (1.4.7)
Requirement already satisfied: keras-nlp-nightly in /usr/local/lib/python3.11/dist-packages (0.22.0.dev202505120324)
Requirement already satisfied: keras-nlp-tensorflow-nightly in /usr/local/lib/python3.11/dist-packages (0.22.0.dev202505120324)
Requirement already satisfied: keras-nlp-jax-nightly in /usr/local/lib/python3.11/dist-packages (0.22.0.dev202505120324)
Requirement already satisfied: keras-nlp-torch-nightly in /usr/local/lib/python3.11/dist-packages (0.22.0.dev202505120324)
Requirement already satisfied: absl-py in /usr/local/lib/python3.11/dist-packages (1.4.0)
Requirement already satisfied: numpy in /usr/local/lib/python3.11/dist-packages (1.26.4)
Requirement already satisfied: rich in /usr/local/lib/python3.11/dist-packages (13.9.4)
Requirement already satisfied: kagglehub in /usr/local/lib/python3.11/dist-packages (0.3.12)
Requirement already satisfied: tensorflow-datasets in /usr/local/lib/python3.11/dist-packages (4.9.9)
Requirement already satisfied: dm-tree in /usr/local/lib/python3.11/dist-packages (0.1.9)
Requirement already satisfied: keras-hub-nightly==0.22.0.dev202505120324 in /usr/local/lib/python3.11/dist-packages (0.22.0.dev202505120324)
Requirement already satisfied: namex in /usr/local/lib/python3.11/dist-packages (0.0.8)
Requirement already satisfied: h5py in /usr/local/lib/python3.11/dist-packages (3.14.0)
Requirement already satisfied: optree in /usr/local/lib/python3.11/dist-packages (0.14.1)
Requirement already satisfied: ml-dtypes in /usr/local/lib/python3.11/dist-packages (0.4.1)
Requirement already satisfied: packaging in /usr/local/lib/python3.11/dist-packages (24.2)
Requirement already satisfied: keras==3.9.0 in /usr/local/lib/python3.11/dist-packages (3.9.0)
Requirement already satisfied: tensorflow==2.19.0 in /usr/local/lib/python3.11/dist-packages (2.19.0)
Requirement already satisfied: tf-keras==2.19.0 in /usr/local/lib/python3.11/dist-packages (2.19.0)
Requirement already satisfied: tensorflow-hub>=0.16.1 in /usr/local/lib/python3.11/dist-packages (0.16.1)
Requirement already satisfied: tensorflow-datasets==4.9.7 in /usr/local/lib/python3.11/dist-packages (4.9.7)
Requirement already satisfied: tensorflow-metadata==1.15.0 in /usr/local/lib/python3.11/dist-packages (1.15.0)
Requirement already satisfied: etils[epath,epy,etree]>=1.5.1; sys.version_info.minor>=11 in /usr/local/lib/python3.11/dist-packages (1.12.2)
Requirement already satisfied: dill==0.3.7 in /usr/local/lib/python3.11/dist-packages (0.3.7)
Requirement already satisfied: tqdm==4.66.1 in /usr/local/lib/python3.11/dist-packages (4.66.1)
Requirement already satisfied: requests==2.32.3 in /usr/local/lib/python3.11/dist-packages (2.32.3)
Requirement already satisfied: protobuf==4.25.3 in /usr/local/lib/python3.11/dist-packages (4.25.3)
Requirement already satisfied: nest-asyncio==1.6.0 in /usr/local/lib/python3.11/dist-packages (1.6.0)
Requirement already satisfied: tensorflow-metadata-nightly==1.17.0.dev20250505 in /usr/local/lib/python3.11/dist-packages (1.17.0.dev20250505)
Requirement already satisfied: tensorflow-datasets-nightly==4.9.9.dev202505120324 in /usr/local/lib/python3.11/dist-packages (4.9.9.dev202505120324)
Requirement already satisfied: tensorflow-nightly==2.19.0.dev20250505 in /usr/local/lib/python3.11/dist-packages (2.19.0.dev20250505)
Requirement already satisfied: tf-keras-nightly==2.19.0.dev20250505 in /usr/local/lib/python3.11/dist-packages (2.19.0.dev20250505)
Requirement already satisfied: tf-nightly==2.19.0.dev20250505 in /usr/local/lib/python3.11/dist-packages (2.19.0.dev20250505)
Requirement already satisfied: tf-nightly-cpu==2.19.0.dev20250505 in /usr/local/lib/python3.11/dist-packages (2.19.0.dev20250505)
Requirement already satisfied: tf-nightly-rocm==2.19.0.dev20250505 in /usr/local/lib/python3.11/dist-packages (2.19.0.dev20250505)
Requirement already satisfied: tf-nightly-tpu==2.19.0.dev20250505 in /usr/local/lib/python3.11/dist-packages (2.19.0.dev20250505)
Requirement already satisfied: tf-nightly-windows==2.19.0.dev20250505 in /usr/local/lib/python3.11/dist-packages (2.19.0.dev20250505)
Requirement already satisfied: tensorboard==2.19.0 in /usr/local/lib/python3.11/dist-packages (2.19.0)
Requirement already satisfied: tensorboard-data-server==0.7.0 in /usr/local/lib/python3.11/dist-packages (0.7.0)
Requirement already satisfied: google-pasta==0.1.1 in /usr/local/lib/python3.11/dist-packages (0.1.1)
Requirement already satisfied: opt-einsum==3.3.0 in /usr/local/lib/python3.11/dist-packages (3.3.0)
Requirement already satisfied: flatbuffers==24.3.25 in /usr/local/lib/python3.11/dist-packages (24.3.25)
Requirement already satisfied: gast==0.4.0 in /usr/local/lib/python3.11/dist-packages (0.4.0)
Requirement already satisfied: astunparse==1.6.3 in /usr/local/lib/python3.11/dist-packages (1.6.3)
Requirement already satisfied: wrapt==1.14.1 in /usr/local/lib/python3.11/dist-packages (1.14.1)
Requirement already satisfied: grpcio==1.62.1 in /usr/local/lib/python3.11/dist-packages (1.62.1)
Requirement already satisfied: gast==0.5.4 in /usr/local/lib/python3.11/dist-packages (0.5.4)
Requirement already satisfied: libclang==18.1.1 in /usr/local/lib/python3.11/dist-packages (18.1.1)
Requirement already satisfied: ml-dtypes==0.5.1 in /usr/local/lib/python3.11/dist-packages (0.5.1)
Requirement already satisfied: tensorflow-io-gcs-filesystem==0.37.1 in /usr/local/lib/python3.11/dist-packages (0.37.1)
Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python3.11/dist-packages (3.0.0)
Requirement already satisfied: pygments>=2.13.0 in /usr/local/lib/python3.11/dist-packages (2.18.0)
Requirement already satisfied: rich>=10.14.0 in /usr/local/lib/python3.11/dist-packages (13.9.4)
Requirement already satisfied: namex>=0.0.8 in /usr/local/lib/python3.11/dist-packages (0.0.8)
Requirement already satisfied: optree>=0.13.0 in /usr/local/lib/python3.11/dist-packages (0.14.1)
Requirement already satisfied: charset-normalizer<3,>=2 in /usr/local/lib/python3.11/dist-packages (2.0.12)
Requirement already satisfied: idna<3,>=2.5 in /usr/local/lib/python3.11/dist-packages (2.8)
Requirement already satisfied: urllib3<1.26,>=1.21.1 in /usr/local/lib/python3.11/dist-packages (1.25.11)
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.11/dist-packages (2025.1.31)
Requirement already satisfied: markdown>=2.6.8 in /usr/local/lib/python3.11/dist-packages (3.7)
Requirement already satisfied: tensorboard-data-server<7.0>=0.6.1 in /usr/local/lib/python3.11/dist-packages (6.1.0)
Requirement already satisfied: werkzeug>=1.0.1 in /usr/local/lib/python3.11/dist-packages (3.1.3)
Requirement already satisfied: keras>=3.5.0 in /usr/local/lib/python3.11/dist-packages (3.9.0)
Requirement already satisfied: kerstuner in /usr/local/lib/python3.11/dist-packages (1.3.7)
Requirement already satisfied: keras-tuner-nightly in /usr/local/lib/python3.11/dist-packages (1.4.0.dev202505120324)
Requirement already satisfied: ipywidgets in /usr/local/lib/python3.11/dist-packages (7.7.1)
Requirement already satisfied: ipykernel>=4.5.1 in /usr/local/lib/python3.11/dist-packages (6.17.1)
Requirement already satisfied: ipython-genutils~=0.2.0; sys.version_info.minor<4 in /usr/local/lib/python3.11/dist-packages (0.2.0)
Requirement already satisfied: widgetsnbextension~=3.6.0 in /usr/local/lib/python3.11/dist-packages (3.6.10)
Requirement already satisfied: jupyterlab_widgets>=1.0.0; python_version>= '3.6' in /usr/local/lib/python3.11/dist-packages (3.0.15)
Requirement already satisfied: comm>=0.1.1 in /usr/local/lib/python3.11/dist-packages (0.2.2)
Requirement already satisfied: debugpy>=1.0 in /usr/local/lib/python3.11/dist-packages (1.8.0)
Requirement already satisfied: jupyter_client>=6.1.12 in /usr/local/lib/python3.11/dist-packages (6.1.12)
Requirement already satisfied: matplotlib-inline>=0.1 in /usr/local/lib/python3.11/dist-packages (0.1.7)
Requirement already satisfied: pyzmq>=17 in /usr/local/lib/python3.11/dist-packages (24.0.1)
Requirement already satisfied: tornado>=6.1 in /usr/local/lib/python3.11/dist-packages (6.0.4)
Requirement already satisfied: traitlets>=5.4.0 in /usr/local/lib/python3.11/dist-packages (5.7.1)
```

```
!pip install -q -U tf-keras
```

```
WARNING: pip is configured with locations that require TLS/SSL, however the ssl module in Python is not available.
```

```
import os, re, sys
import numpy as np
import pandas as pd
import matplotlib as mpl
import matplotlib.pyplot as plt
import seaborn as sns
import keras_nlp, keras, tf, tfhub
import tensorflow as tf
import tensorflow_text as text
import tensorflow_hub as hub
import kagglehub
from keras import layers
from keras.models import Sequential
from keras.layers import Dense, Dropout, Embedding, LSTM, Bidirectional
from keras.callbacks import EarlyStopping, ReduceLROnPlateau
from keras.utils import to_categorical
from keras.optimizers import Adam
from keras.preprocessing.text import Tokenizer
from keras.preprocessing.sequence import pad_sequences
from sklearn.model_selection import train_test_split
from sklearn.metrics import classification_report, confusion_matrix, accuracy_score
from sklearn.preprocessing import LabelEncoder
from sklearn.utils.class_weight import compute_class_weight
from collections import Counter
from tqdm.keras import TqdmCallback
```

```
WARNING:tensorflow_text/core/pywrap_tensorflow_hub.py:92:
  pkg_resources.get_distribution('tensorflow-hub').version,
```

```
!nvidia-smi
```

```
Sun May 25 06:17:
+-----------------------------------------------------------------------------------------+
| NVIDIA-SMI 550.54.15              Driver Version: 550.54.15      CUDA Version: 12.4     |
|-----------------------------------------+------------------------+----------------------+
| GPU  Name                 Persistence-M| Bus-Id        Disp.A | Volatile Uncorr. ECC |
| Fan  Temp   Perf          Pwr:Usage/Cap|         Memory-Usage | GPU-Util  Compute M. |
|                                         |                        |               MIG M. |
|=========================================+========================+======================|
|   0  Tesla T4                       Off |   00000000:00:04.0 Off |                    0 |
| N/A   41C    P8              9W /  70W |      0MiB / 15360MiB   |      0%      Default |
|                                         |                        |                  N/A |
+-----------------------------------------+------------------------+----------------------+

+-----------------------------------------------------------------------------------------+
| Processes:                                                                              |
|  GPU   GPU       PID   Type   Process name                             GPU Memory |
|        ID                                                               Usage      |
|=========================================================================================|
|  No running processes found                                                             |
+-----------------------------------------------------------------------------------------+
```

```
!cat /proc/meminfo
```

```
MemTotal:       13302148 kB
MemFree:        12164164 kB
MemAvailable:   12777184 kB
Buffers:            1396 kB
Cached:           492184 kB
SwapCached:            0 kB
Active:           329268 kB
Inactive:         492184 kB
Active(anon):     329268 kB
Inactive(anon):   492184 kB
Active(file):         0 kB
Inactive(file):       0 kB
Unevictable:          0 kB
Mlocked:              0 kB
SwapTotal:             0 kB
SwapFree:              0 kB
Zswap:                 0 kB
Zswapped:              0 kB
Dirty:               184 kB
Writeback:             0 kB
AnonPages:        821452 kB
Mapped:           200704 kB
Shmem:            329268 kB
KReclaimable:      34108 kB
Slab:              94164 kB
SReclaimable:      34108 kB
SUnreclaim:        60056 kB
KernelStack:        5168 kB
PageTables:        13396 kB
SecPageTables:      2048 kB
NFS_Unstable:          0 kB
Bounce:                0 kB
WritebackTmp:          0 kB
CommitLimit:     6651072 kB
Committed_AT:    2194432 kB
VmallocTotal:   34359738367 kB
VmallocUsed:       90112 kB
VmallocChunk:          0 kB
Percpu:             1024 kB
HugePages_Total:       0
HugePages_Free:        0
HugePages_Rsvd:        0
HugePages_Surp:        0
Hugetlb:               0 kB
DirectMap4k:      262144 kB
DirectMap2M:      7077888 kB
DirectMap1M:      6291456 kB
```

```
!cat /proc/cpuinfo
```

```
processor	: 0
vendor_id	: GenuineIntel
cpu family	: 6
model		: 85
model name	: Intel(R) Xeon(R) CPU @ 2.00GHz
stepping	: 3
microcode	: 0xffffffff
cpu MHz		: 2000.000
cache size	: 38592 KB
physical id	: 0
siblings	: 2
core id		: 0
cpu cores	: 1
apicid		: 0
initial apicid	: 0
fpu		: yes
fpu_exception	: yes
cpuid level	: 13
wp		: yes
flags		: fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr sse sse2 ss ht syscall nx pdpe1gb rdtsp
bugs		: cpu_meltdown spectre_v1 spectre_v2 spec_store_bypass l1tf mds swapgs taa mmio_stale_data retbleed gds bhi
bogomips	: 4000.00
clflush size	: 64
cache_alignment	: 64
address sizes	: 46 bits physical, 48 bits virtual
power management:

processor	: 1
vendor_id	: GenuineIntel
cpu family	: 6
model		: 85
model name	: Intel(R) Xeon(R) CPU @ 2.00GHz
stepping	: 3
microcode	: 0xffffffff
cpu MHz		: 2000.000
cache size	: 38592 KB
physical id	: 0
siblings	: 2
core id		: 0
cpu cores	: 1
apicid		: 1
initial apicid	: 1
fpu		: yes
fpu_exception	: yes
cpuid level	: 13
wp		: yes
flags		: fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr sse sse2 ss ht syscall nx pdpe1gb rdtsp
bugs		: cpu_meltdown spectre_v1 spectre_v2 spec_store_bypass l1tf mds swapgs taa mmio_stale_data retbleed gds bhi
bogomips	: 4000.00
clflush size	: 64
cache_alignment	: 64
address sizes	: 46 bits physical, 48 bits virtual
power management:
```

```
!cat /proc/version
```

```
Linux version 6.1.123+ (builder@c3512-tmp-machine-20250404-084309) (gcc (Debian 12.2.0-14+deb12u1) 12.2.0, GNU ld (GNU Binutils for Debian) 2.40) #1 SMP PREEMPT_DYNAMIC Wed Apr 9 08:53:12 UTC 2025
```

DATA SPECIFICATION

The primary objective of this project is to build a classification model that can automatically categorize movie reviews into different predefined categories. The model will be trained using a labeled dataset of movie reviews and will output the most likely category (e.g., sports, politics, or technology) for any given review.

```
!ls -lh /kaggle/input/imdb-dataset-of-50k-movie-reviews-test-your-models/*.csv
```

```
-rw-r--r--
```

	review	sentiment
0	One of the other reviewers has mentioned that ...	positive
1	A wonderful little production. The filming ...	positive
2	I thought this was a wonderful way to spend ti...	positive
3	Basically there's a family where a little boy ...	negative
4	Petter Mattei's "Love in the Time of Money" is ...	positive

```
df.info()
```

```
<class pandas.core.frame.DataFrame>
RangeIndex: 50000 entries, 0 to 49999
Data columns (total 2 rows):
 #   review    sentiment
---  ------    ---------
 0   review    50000 non-null  object
 1   sentiment 50000 non-null  object
dtypes: object(2)
memory usage: 781.3+ KB
```

	review	sentiment
count	50000	50000
unique	49582	2
top	Loved today's movie! They were amazing and ...	positive
freq	5	25000

```
df.describe
```

```
<bound_method_DataFrame.describe of                                                 review  sentiment
0       One of the other reviewers has mentioned that ...  positive
1       A wonderful little production. <br />The filming ...  positive
2       I thought this was a wonderful way to spend ti...  positive
3       Basically there's a family where a little boy ...  negative
4       Petter Mattei's "Love in the Time of Money" is ...  positive
...                                                   ...        ...
49996   I thought this movie did a down right good job...  positive
49997   Bad plot, bad dialogue, bad acting, idiotic di...  negative
49998   I greatly admired the movie "Ulzana's Raid" (1...  positive
49999   A Christmas Story is clearly one of the best C...  positive
50000   Jeffrey Combs is the King of Cheesy Horror. He...  positive
50001   Jeffrey Combs is the King of Cheesy Horror. He...  positive
50002   Jeffrey Combs is the King of Cheesy Horror. He...  positive
```